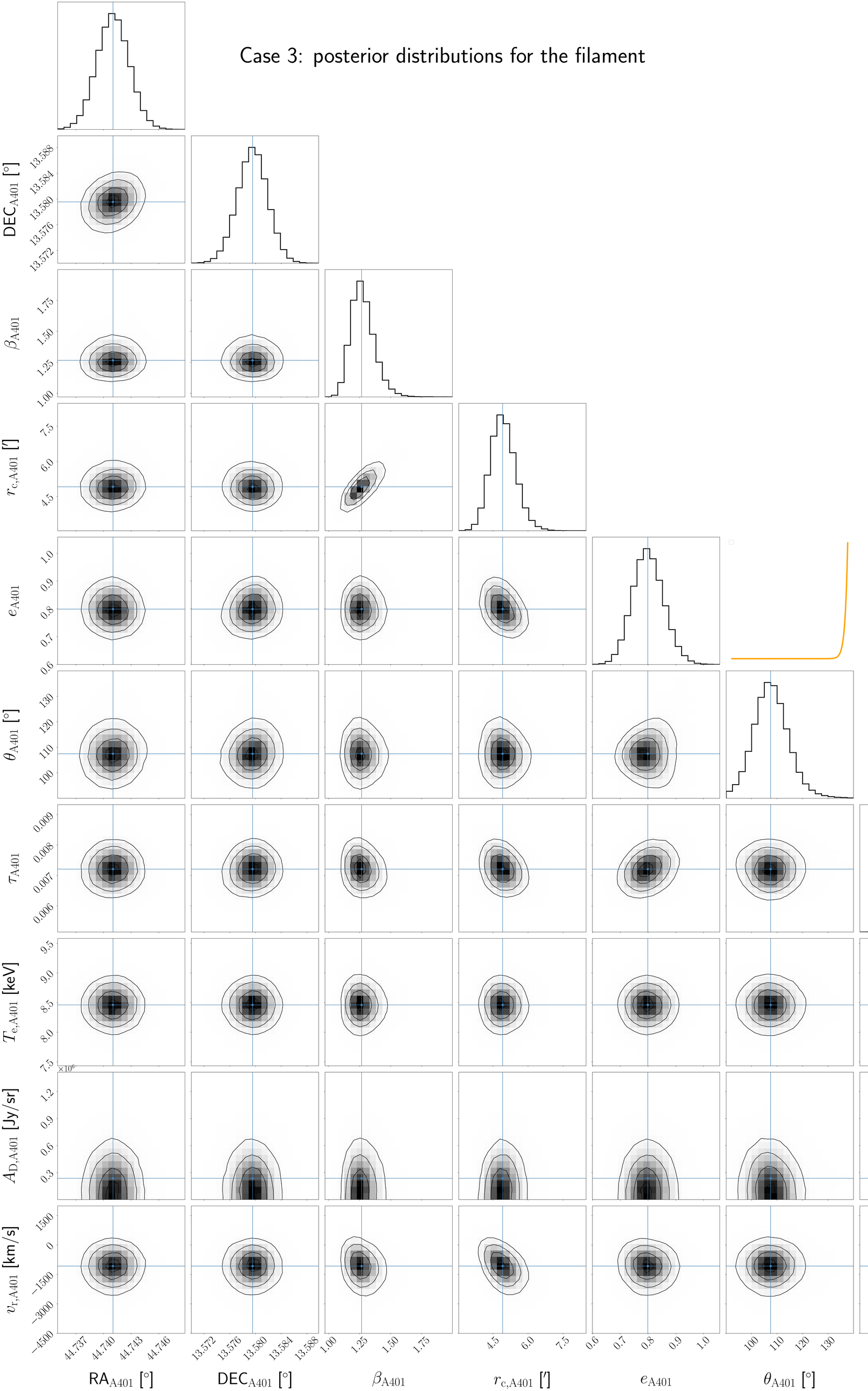Case 3: posterior distributions for the filament